

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03 COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to

minimize
waiting time and improve service quality.

(5 Marks)

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations.

(5 Marks)

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

(5 Marks)

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance.

(5 Marks)

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

(5 Marks)

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes.

(5 Marks)

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

(5 Marks)

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

(5 Marks)

9. An online advertising system uses Reinforcement Learning to maximize clickthrough rates. Outline reward design challenges and explain their impact on system performance.

(5 Marks)

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function.

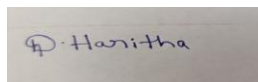
(5 Marks)

Code of Conduct:

I Dasari Haritha-192425254 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CO2 AT2

APPLICATION - BASED QUESTIONS

Question 1

Design and develop a Reinforcement Learning framework for a ride-sharing company that uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality.

Answer

Objective

The objective is to match passengers with the most suitable driver by reducing passenger waiting time, minimizing travel distance, improving driver utilization, and increasing customer satisfaction.

Why Reinforcement Learning?

Reinforcement Learning (RL) is suitable because passenger requests, traffic conditions, and driver availability change continuously. The RL agent learns from previous decisions and improves driver allocation over time.

RL Framework

Agent: Ride-sharing management system

Environment: Passengers, drivers, roads, traffic conditions, and GPS data.

State Space

The state contains:

- Driver location
- Passenger location
- Driver availability
- Traffic conditions
- Distance between driver and passenger

- Time of day
- Ride demand

Action Space

The agent can:

- Assign the nearest driver
- Assign another available driver
- Reassign a driver
- Reject the request if no driver is available
- Delay assignment for a better match

Reward Function

Action	Reward
Passenger picked up quickly	+40
Successful trip completed	+50
Short travel distance	+20
High customer rating	+20
Long waiting time	−25
Ride cancellation	−40
Wrong driver assignment	−30

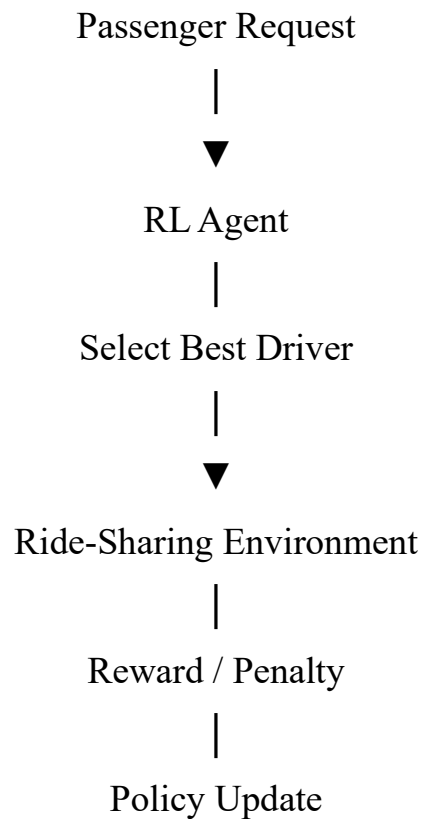
Policy

The policy selects the driver that gives the maximum long-term reward by considering waiting time, traffic, and driver distance.

Learning Process

The RL agent first explores different driver assignments and later exploits the best-performing strategies. After every completed trip, it updates its policy using the received rewards.

RL Architecture



Expected Outcomes

- Reduced passenger waiting time
- Better driver utilization
- Lower fuel consumption
- Improved customer satisfaction
- Higher company revenue

Conclusion

Reinforcement Learning continuously improves driver-passenger matching by learning from real-time interactions, resulting in efficient transportation services.

Question 2

An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations.

Answer

Objective

The objective is to recommend personalized study materials that improve learning performance and keep students engaged.

Why Reinforcement Learning?

Every student learns differently. RL adapts recommendations based on student performance, interests, and feedback instead of using fixed rules.

RL Framework

Agent: Recommendation system

Environment: Students, learning platform, quizzes, and study materials.

State Space

- Student performance
- Previous lessons completed
- Quiz scores
- Learning speed
- Interests
- Time spent studying

Action Space

- Recommend familiar topics
- Recommend advanced topics
- Suggest practice questions
- Suggest revision materials
- Recommend videos

Reward Function

Situation	Reward
Lesson completed	+20
High quiz score	+30
Positive student feedback	+20
Student skips lesson	-10
Low quiz score	-15

Exploration vs Exploitation

Exploration: The system recommends new topics or learning resources to discover the student's interests.

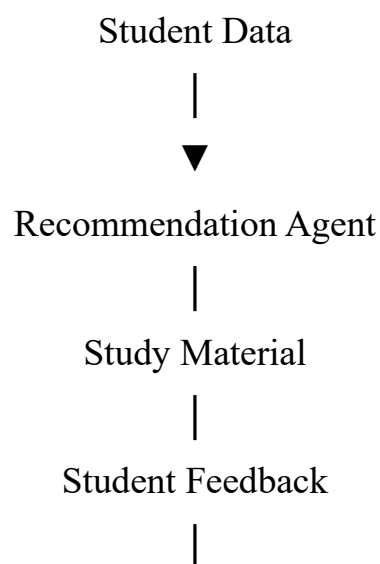
Exploitation: The system recommends materials similar to those that previously helped the student perform well.

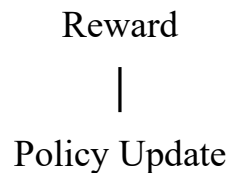
A balance between exploration and exploitation provides better long-term learning.

Learning Process

The RL agent updates its recommendation policy after every quiz, assignment, or student interaction.

RL Architecture





Expected Outcomes

- Personalized learning
- Better academic performance
- Higher student engagement
- Improved learning efficiency

Conclusion

Reinforcement Learning creates an adaptive learning environment by continuously improving recommendations based on student performance.

Question 3

Design and develop a Reinforcement Learning framework for a robotic vacuum cleaner that cleans efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

Answer

Objective

The objective is to clean the maximum area efficiently while avoiding obstacles and minimizing battery usage.

Why Reinforcement Learning?

The environment changes as furniture or obstacles move. RL enables the robot to learn efficient cleaning paths through experience.

RL Framework

Agent: Robotic vacuum cleaner

Environment: Rooms, furniture, walls, obstacles, and charging station.

State Space

- Robot location
- Battery level
- Obstacle position
- Cleaned area
- Dirt level
- Charging station location

Action Space

- Move forward
- Move backward
- Turn left
- Turn right
- Clean
- Return to charging station

Reward Function

Situation	Reward
Clean new area	+20
Finish cleaning	+50
Save battery	+15
Hit obstacle	−30
Repeat same area	−10

Policy

The policy selects actions that maximize cleaning efficiency while avoiding unnecessary movements and collisions.

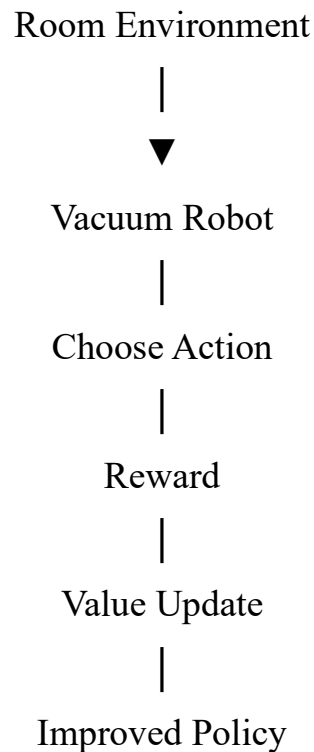
Value Function

The value function estimates the future reward of each state. States leading to more cleaned areas and fewer collisions receive higher values. Over time, the robot prefers these high-value states and improves cleaning performance.

Learning Process

The robot learns from every cleaning cycle and gradually discovers the most efficient cleaning path.

RL Architecture



Expected Outcomes

- Faster cleaning
- Better battery usage
- Fewer collisions
- Improved cleaning efficiency

Conclusion

Using RL, the robotic vacuum continuously learns better cleaning strategies, resulting in improved efficiency and reduced energy consumption.

Question 4

A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance.

Answer

Objective

The objective is to maximize long-term profit while reducing trading risk.

Why Reinforcement Learning?

Cryptocurrency prices change rapidly. RL learns profitable trading strategies by interacting with market conditions.

RL Framework

Agent: Trading system

Environment: Cryptocurrency market.

State Space

- Current market price
- Price history
- Trading volume
- Market trend
- Portfolio balance
- Risk level

Action Space

- Buy
- Sell
- Hold

Reward Function

Situation	Reward
Profitable trade	+50

Situation	Reward
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Long-term profit	+30
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Small profit	+10
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Financial loss	-40
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High risk decision	-20
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Challenges in Reward Design

- Markets are highly volatile.
- Profit may occur after several trades (delayed rewards).
- A reward focused only on profit may encourage risky trading.
- Poor reward design can lead to frequent unnecessary trades or large financial losses.

Learning Process

The RL agent updates its strategy after each trade and gradually learns profitable buying and selling decisions.

RL Architecture



Expected Outcomes

- Better investment decisions
- Reduced trading losses
- Improved long-term profit
- Smarter risk management

Conclusion

A carefully designed reward function is essential for RL-based trading systems because it guides the agent toward profitable and low-risk decisions.

Question 5

Develop a smart grid system that uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

Answer

Objective

The objective is to distribute electricity efficiently, reduce power wastage, and maintain a stable electricity supply.

Why Reinforcement Learning?

Electricity demand changes throughout the day. RL learns the best distribution strategy by observing demand and supply patterns.

RL Framework

Agent: Smart grid controller

Environment: Power stations, consumers, renewable energy sources, and electricity network.

State Space

- Electricity demand
- Power generation
- Battery storage level

- Renewable energy availability
- Peak load
- Grid condition

Action Space

- Increase power supply
- Reduce power supply
- Store excess energy
- Release stored energy
- Balance load between regions

Reward Function

Situation	Reward
Stable electricity supply	+50
Reduced energy wastage	+20
Efficient load balancing	+20
Power outage	-50
Overloaded grid	-30
Energy wastage	-20

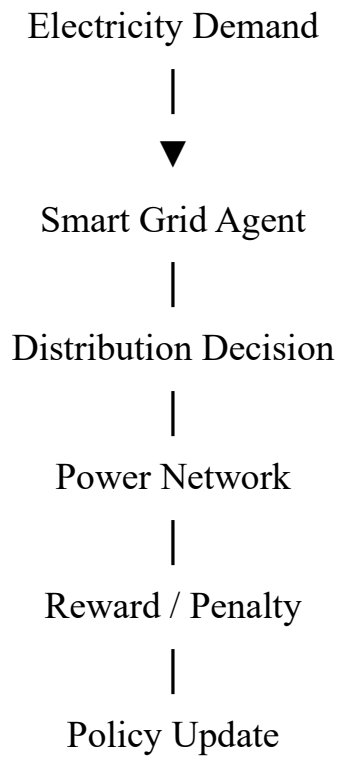
Policy

The policy selects the action that provides a reliable electricity supply while minimizing operational costs and energy waste.

Learning Process

The RL agent continuously learns from electricity demand and updates its policy to improve future distribution decisions.

RL Architecture



Expected Outcomes

- Reduced power outages
- Better energy efficiency
- Improved renewable energy utilization
- Lower operational costs
- Reliable electricity distribution

Conclusion

Reinforcement Learning enables smart grids to adapt to changing electricity demand, improving energy efficiency, grid reliability, and overall system performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, wellstructured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, wellorganized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand